



**BANGLADESH UNIVERSITY OF  
ENGINEERING AND TECHNOLOGY  
DEPARTMENT OF MECHANICAL ENGINEERING**

**PROJECT REPORT**

**PROJECT NAME:**

**Modeling CNC Machine in  
SOLIDWORKS**

**Course no : ME 260**

**Course name : MECHANICAL ENGINEERING  
DRAWING II**

**Members:**

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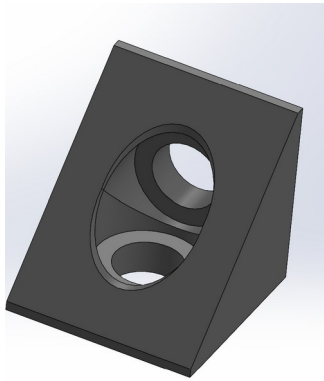
**ID: 2310083 – Tasfiya Alam**

**ID: 2310084 – Md. Al Amin**

**ID: 2310085 – Md. Sifat Ibn Azad**

## Part 1: Black Angle Corner Connector

Material: 6063-T5 Aluminium Alloy (black electrophoretic coating)



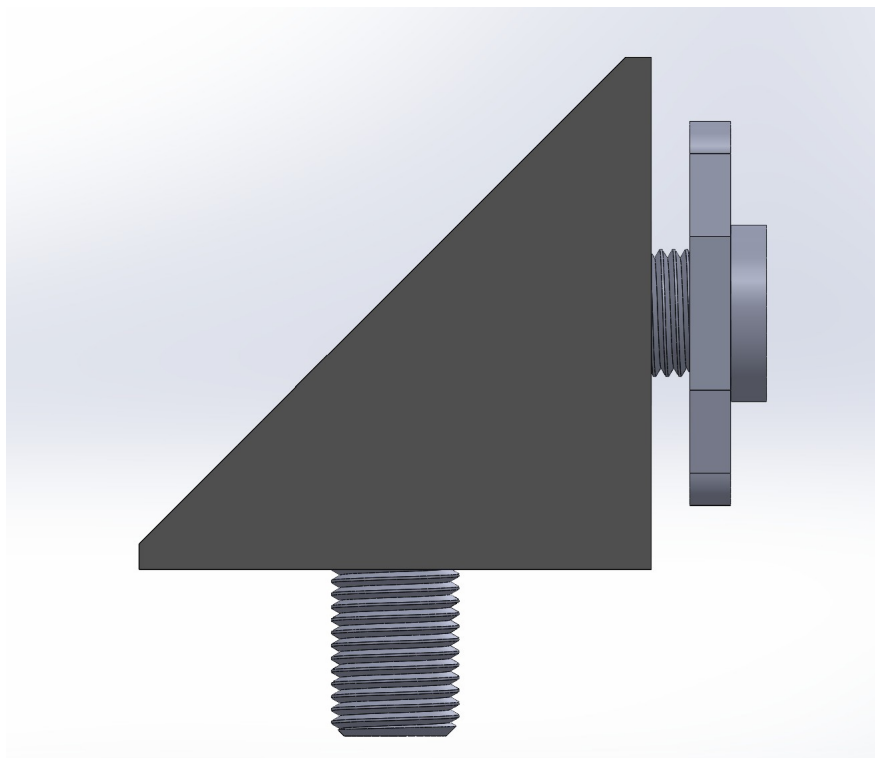
Mass = 8.67 grams

**Material Justification:** 6063-T5 aluminium is light, corrosion-resistant and easy to form, and it matches the aluminium profiles it joins, so there is no galvanic corrosion at the joint. The black electrophoretic coating adds a hard, wear- and rust-resistant finish.

### Cost Analysis (in BDT)

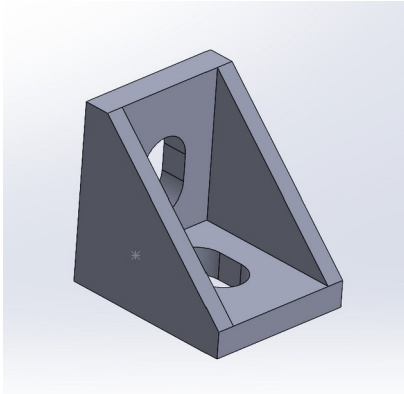
| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate    | Total Cost (Qty 1) |
|-----------|---------------|--------------------------------|--------------------|
| 0.00867   | 560.00        | $0.00867 \times 560.00 = 4.86$ | 4.86               |

## Assembly: Black Angle Corner (After Assembly)

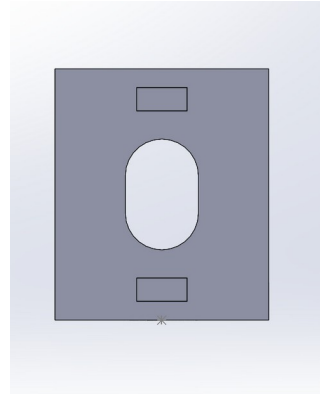


## Part 2: Cast Corner Bracket

Material: Die-Cast Aluminium



Mass = 2.22  
grams



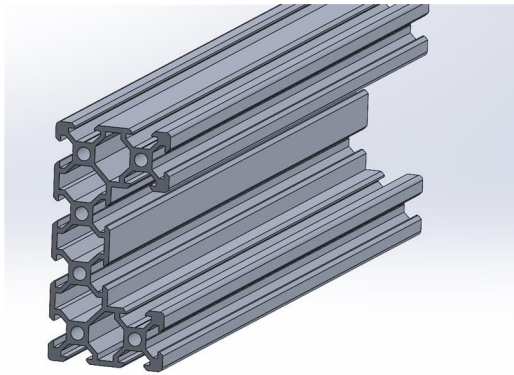
**Material Justification:** Die casting produces the ribbed bracket shape in a single step with good dimensional accuracy and low unit cost. Aluminium keeps the frame corners light, rigid and corrosion-resistant, and compatible with the aluminium extrusions.

### Cost Analysis (in BDT)

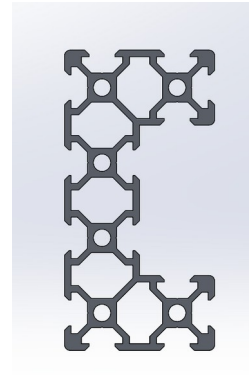
| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate    | Total Cost (Qty 1) |
|-----------|---------------|--------------------------------|--------------------|
| 0.00222   | 450.00        | $0.00222 \times 450.00 = 1.00$ | 1.00               |

## Part 3: C-Beam Linear Rail (40×80×250 mm)

Material: Extruded 6063-T5 Aluminium Alloy



Mass = 539.41  
grams



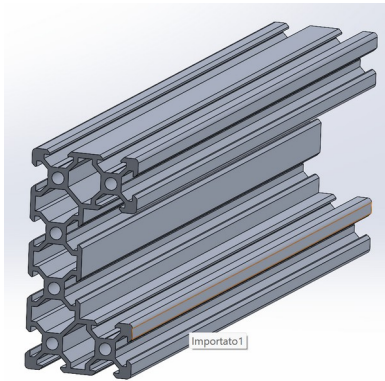
**Material Justification:** 6063-T5 is the standard extrusion alloy and can be pushed into the complex C-beam profile with a smooth surface that V-wheels run on accurately. It gives enough stiffness for this short member while keeping the frame light and rust-free.

### Cost Analysis (in BDT)

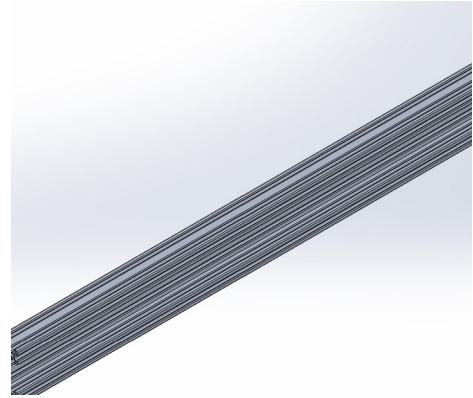
| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate      | Total Cost (Qty 1) |
|-----------|---------------|----------------------------------|--------------------|
| 0.53941   | 560.00        | $0.53941 \times 560.00 = 302.07$ | 302.07             |

## Part 4: C-Beam Linear Rail (40×80×1000 mm)

Material: Extruded 6063-T5 Aluminium Alloy



Mass =  
2,157.64 grams



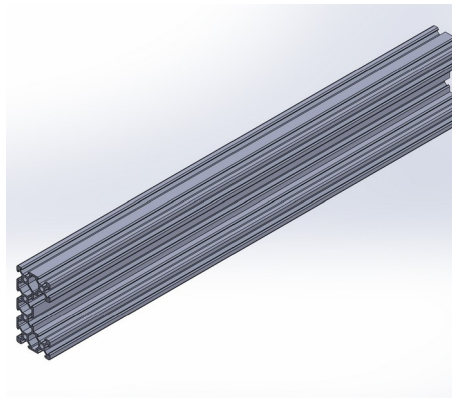
**Material Justification:** As the longest rail, this member needs high bending stiffness at low weight. Extruded 6063-T5 gives a stiff, straight profile with a smooth running surface for the V-wheels and natural corrosion resistance, at a low cost per kg.

### Cost Analysis (in BDT)

| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate        | Total Cost (Qty 1) |
|-----------|---------------|------------------------------------|--------------------|
| 2.15764   | 560.00        | $2.15764 \times 560.00 = 1,208.28$ | 1,208.28           |

## Part 5: C-Beam Linear Rail (40×80×500 mm)

Material: Extruded 6063-T5 Aluminium Alloy



Mass = 1,078.82  
grams

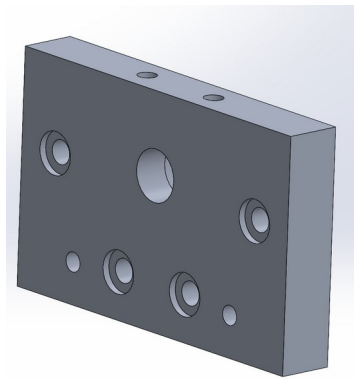
**Material Justification:** Extruded 6063-T5 gives a light yet stiff linear rail with precise V-grooves for smooth wheel travel. Using the same alloy as the other rails keeps thermal expansion uniform across the machine and simplifies sourcing.

### Cost Analysis (in BDT)

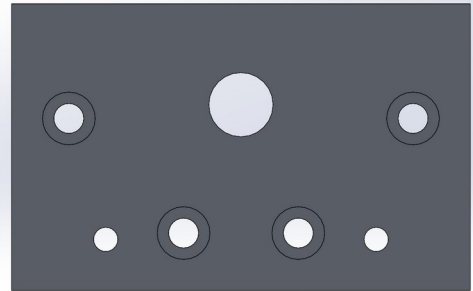
| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate      | Total Cost (Qty 1) |
|-----------|---------------|----------------------------------|--------------------|
| 1.07882   | 560.00        | $1.07882 \times 560.00 = 604.14$ | 604.14             |

## Part 6: C-Beam End Mount

Material: 6063-T5 Aluminium Alloy



Mass = 149.65  
grams



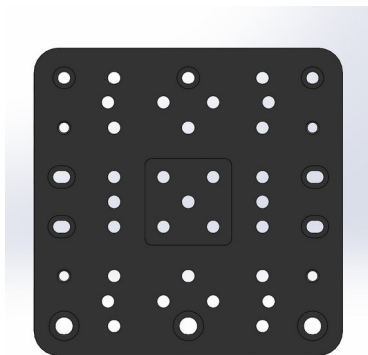
**Material Justification:** The end mount carries the motor and lead-screw bearing loads, so it must be rigid and machined accurately. 6063-T5 is easy to drill and counterbore, light, and matches the C-beam, avoiding galvanic corrosion and thermal mismatch.

### Cost Analysis (in BDT)

| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate     | Total Cost (Qty 1) |
|-----------|---------------|---------------------------------|--------------------|
| 0.14965   | 560.00        | $0.14965 \times 560.00 = 83.80$ | 83.80              |

## Part 7: Gantry Plate

Material: 6061-T5 Aluminium Alloy (black anodized)



Mass = 228.57 grams

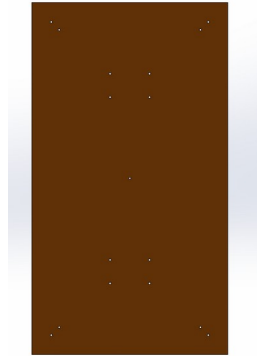
**Material Justification:** 6061-T5 has higher strength than 6063, so the plate stays flat and holds the V-wheel bolts in position under cutting loads. It machines cleanly, and black anodizing gives a hard, wear- and corrosion-resistant surface.

### Cost Analysis (in BDT)

| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate      | Total Cost (Qty 1) |
|-----------|---------------|----------------------------------|--------------------|
| 0.22857   | 620.00        | $0.22857 \times 620.00 = 141.71$ | 141.71             |

## Part 8: MDF Board (900×500×14 mm)

Material: MDF – Fine Wood Fibres (Mahogany)



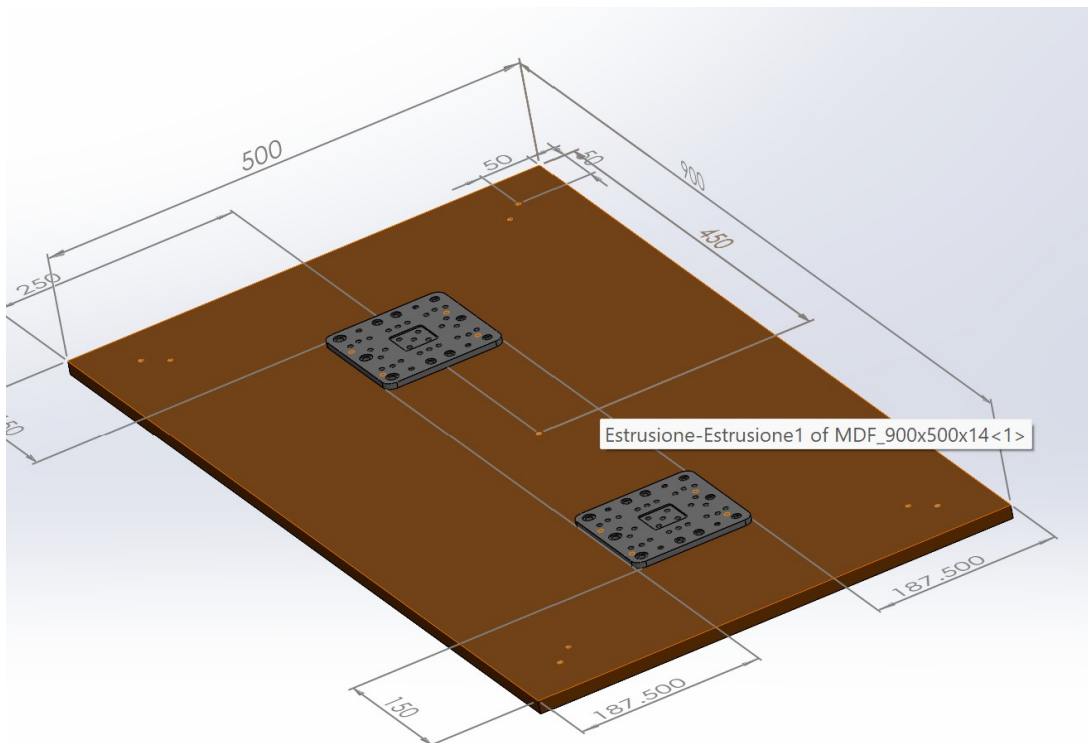
Mass = 3,777.02  
grams

**Material Justification:** MDF is flat, dimensionally stable and inexpensive, and it is easy to drill and screw into, which suits a machine bed. Its fibrous structure damps vibration, and it is widely available in Bangladesh and easy to replace when worn.

### Cost Analysis (in BDT)

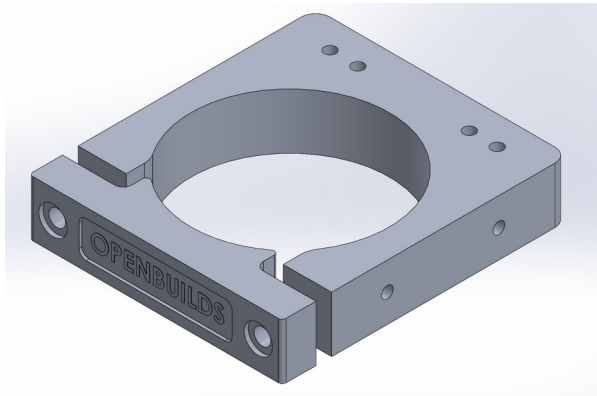
| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate      | Total Cost (Qty 1) |
|-----------|---------------|----------------------------------|--------------------|
| 3.77702   | 100.11        | $3.77702 \times 100.11 = 378.12$ | 378.12             |

## Assembly: Gantry Plate & MDF Board (After Assembly)



## Part 9: Router Spindle Mount

Material: 6063-T5 Aluminium Alloy



Mass = 262.19 grams

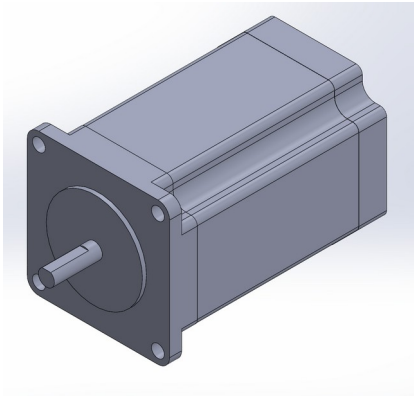
**Material Justification:** The mount must clamp the spindle rigidly and carry cutting forces without adding much weight to the moving gantry. 6063-T5 aluminium provides good rigidity for its weight, conducts heat away from the spindle and machines easily.

### Cost Analysis (in BDT)

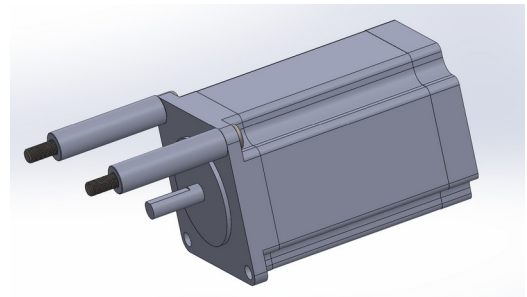
| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate      | Total Cost (Qty 1) |
|-----------|---------------|----------------------------------|--------------------|
| 0.26219   | 560.00        | $0.26219 \times 560.00 = 146.83$ | 146.83             |

## Part 10: NEMA 23 Stepper Motor

Material: Aluminium alloy housing (bought-out part)



Mass = 750 –  
900 g



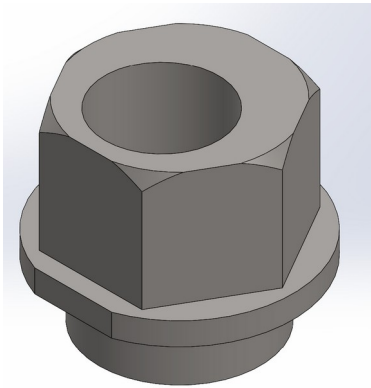
**Material Justification:** The aluminium housing is light and dissipates heat from the windings. Copper windings give high conductivity, silicon-steel laminations reduce eddy-current losses and the hardened steel shaft resists wear and bending.

### Cost Analysis (in BDT)

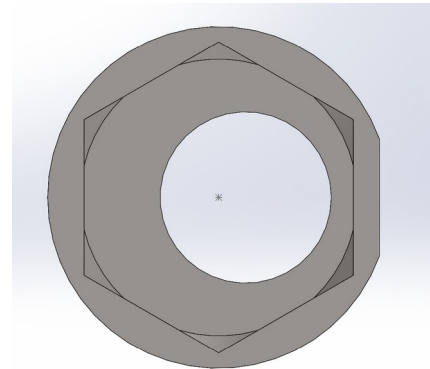
| Item                  | Unit Price | Quantity | Total Cost |
|-----------------------|------------|----------|------------|
| NEMA 23 Stepper Motor | 4,900.00   | 1        | 4,900.00   |

## Part 11: Eccentric Spacer

Material: AISI 304 Stainless Steel



Mass = 2.22  
grams



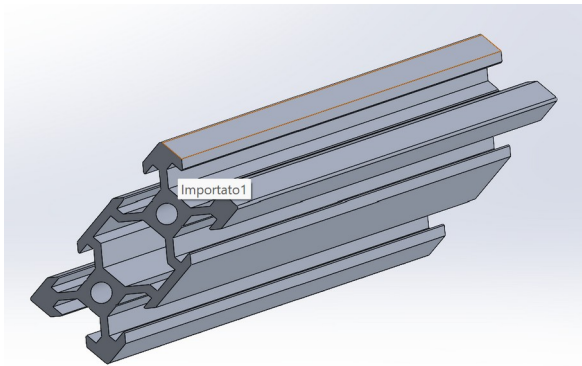
**Material Justification:** The spacer holds the wheel axle off-centre under preload and repeated adjustment, so it must not deform when tightened. AISI 304 stainless steel is strong, tough and corrosion-resistant, and the small part size keeps the cost negligible.

### Cost Analysis (in BDT)

| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate    | Total Cost (Qty 1) |
|-----------|---------------|--------------------------------|--------------------|
| 0.00222   | 450.00        | $0.00222 \times 450.00 = 1.00$ | 1.00               |

## Part 12: V-Slot Profile (20×40×960 mm)

Material: Extruded 6063-T5 Aluminium Alloy



Mass = 777.57 grams

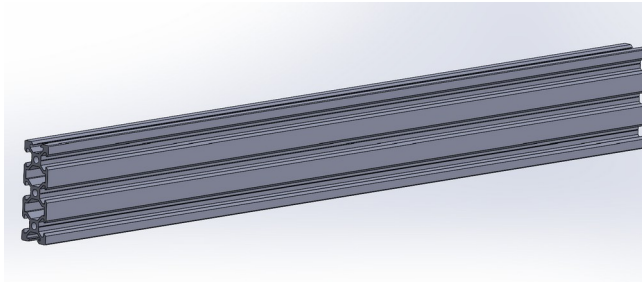
**Material Justification:** Extruded 6063-T5 forms the V-slot profile accurately, giving straight V-grooves for the wheels and T-slots for flexible mounting. It is light, corrosion-resistant and cheap per kg, which makes it ideal for structural members.

### Cost Analysis (in BDT)

| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate      | Total Cost (Qty 1) |
|-----------|---------------|----------------------------------|--------------------|
| 0.77757   | 560.00        | $0.77757 \times 560.00 = 435.44$ | 435.44             |

### Part 13: V-Slot Profile (20×60×500 mm)

Material: Extruded 6063-T5 Aluminium Alloy



Mass = 578.58 grams

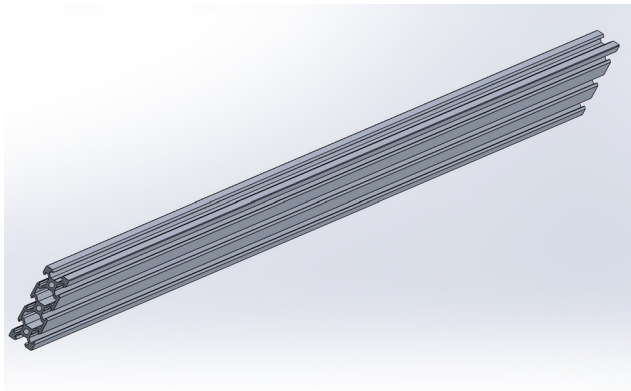
**Material Justification:** 6063-T5 is well suited to extrusion, so the 20×60 profile is produced with precise grooves and a good surface finish. It offers adequate stiffness for this short member at low weight and does not need painting.

#### Cost Analysis (in BDT)

| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate      | Total Cost (Qty 1) |
|-----------|---------------|----------------------------------|--------------------|
| 0.57858   | 560.00        | $0.57858 \times 560.00 = 324.00$ | 324.00             |

### Part 14: V-Slot Profile (20×60×1000 mm)

Material: Extruded 6063-T5 Aluminium Alloy



Mass = 1,157.16  
grams

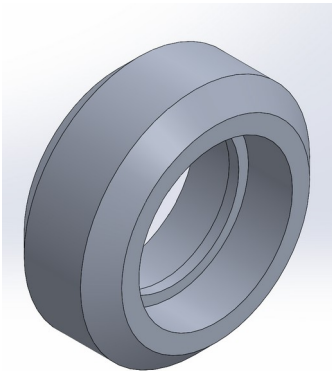
**Material Justification:** The 1000 mm length needs good bending stiffness over a long span. The wide 20×60 extruded 6063-T5 section provides this at low weight, with corrosion resistance and precise V-grooves for smooth motion.

#### Cost Analysis (in BDT)

| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate      | Total Cost (Qty 1) |
|-----------|---------------|----------------------------------|--------------------|
| 1.15716   | 560.00        | $1.15716 \times 560.00 = 648.01$ | 648.01             |

# Part 15: Xtreme Solid V-Wheel

Material: 6063-T5 Aluminium Alloy



Mass = 6.06 grams

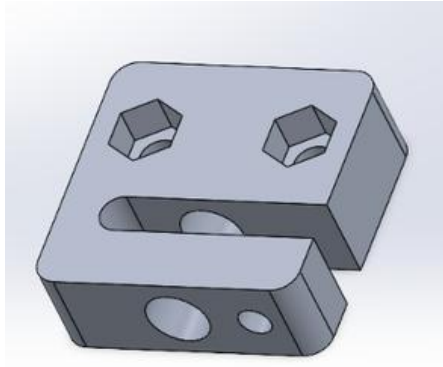
**Material Justification:** Aluminium gives the wheel a dimensionally stable, wear-resistant rim that keeps its shape under preload on the V-groove. Using the same alloy as the rail means both expand equally with temperature, so the wheel fit stays consistent.

## Cost Analysis (in BDT)

| Mass (kg) | Rate (BDT/kg) | Material Cost = Mass × Rate    | Total Cost (Qty 1) |
|-----------|---------------|--------------------------------|--------------------|
| 0.00606   | 560.00        | $0.00606 \times 560.00 = 3.39$ | 3.39               |

## Part 16: 8mm Acme Anti Backlash Nut Block

Material : Delrin(POM)



**Mass : 12.92 grams**

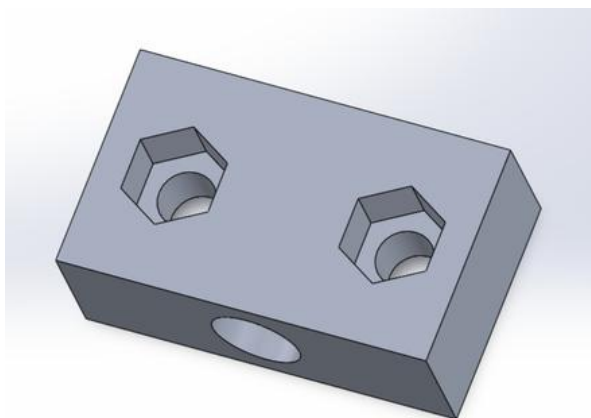
**Material Justification :** Selected for its high hardness and strength, which helps resist thread wear and deformation under repeated sliding and preload forces. It ensures accurate motion and longer service life.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.01292  | 800          | 10.34         | 75.85             |

## Part 17 : 8mm Acme Nut Block

Material : Delrin(POM)



**Mass : 8.66 grams**

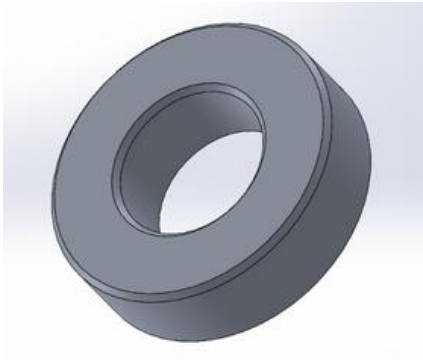
**Material Justification :** because of its excellent strength, hardness, and wear resistance. It helps maintain thread accuracy and withstand repeated motion and preload forces without significant deformation.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00866  | 800          | 6.93          | 67.33             |

## Part 18 : 3 mm Aliminum spacer

Material : 6063-T5 Aluminum Alloy



**Mass : 0.47 gram**

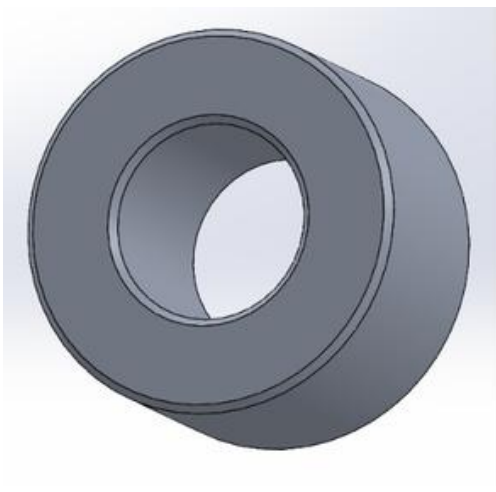
**Material Justification :** Used as a spacer because it requires a lightweight material with good dimensional accuracy and corrosion resistance. 6063 alloy is commonly used for structural and extrusion parts where smooth surface finish is needed.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00047  | 600          | 0.28          | 50.70             |

## Part 19 : 6 mm Aliminum spacer

Material : 6063-T5 Aluminum Alloy



**Mass : 0.94 gram**

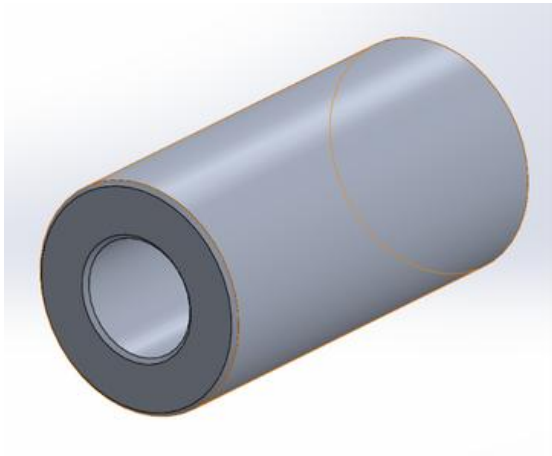
**Material Justification :** 6063-T5 aluminum alloy is used for the spacer because it provides a lightweight structure with excellent corrosion resistance and maintains accurate dimensions. Its good surface finish and extrusion capability make it suitable for precision spacer applications.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00094  | 600          | 0.56          | 51.40             |

## Part 20 : 20 mm Aliminum spacer

Material : 6063-T5 Aluminum Alloy



**Mass : 3.13 gram**

**Material Justification :** The spacer is made from 6063-T5 aluminum alloy due to its low weight, good dimensional stability, and resistance to environmental corrosion. This alloy is widely preferred for components requiring smooth finishing and reliable structural performance.

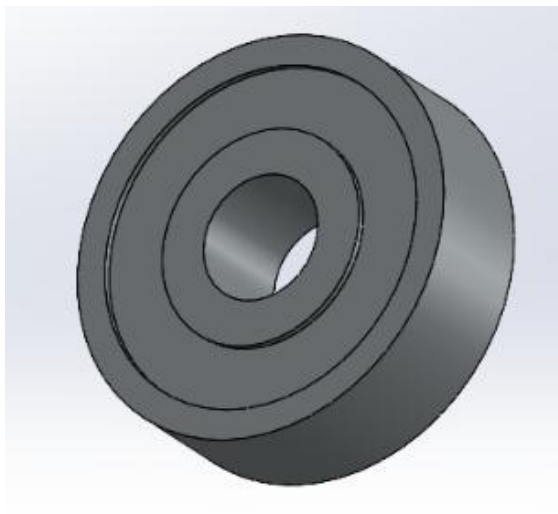
**Cost analysis in**

**BDT**

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00313  | 600          | 1.88          | 54.70             |

## Part 21 : Ball bearing 5x16x5

Material : Chrome stainless steel



**Mass : 6.67 grams**

**Material justification :** Ball bearings require a material that can withstand repeated contact and friction during rotation. Chrome stainless steel is selected due to its high hardness, strength, and resistance to wear, which helps maintain reliable performance over a long period.

**Cost analysis in BDT**

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00667  | 1200         | 8.00          | 70.00             |

## Part 22: Ball bearing 8x16x5

Material : Chrome stainless steel



**Mass : 5.50 grams**

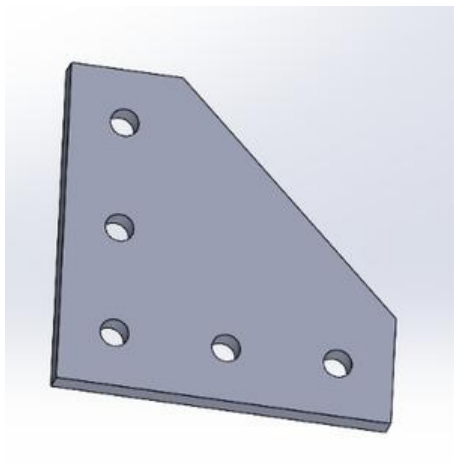
**Material justification:** Ball bearings experience continuous rolling contact and high friction. Chrome stainless steel is selected because it provides high hardness and wear resistance, allowing smooth rotation and long service life

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.0055   | 1200         | 6.60          | 66.50             |

## Part 23 : joining plate 90 degree

Material : 6061-T5 Aluminum Alloy



**Mass : 29.14 grams**

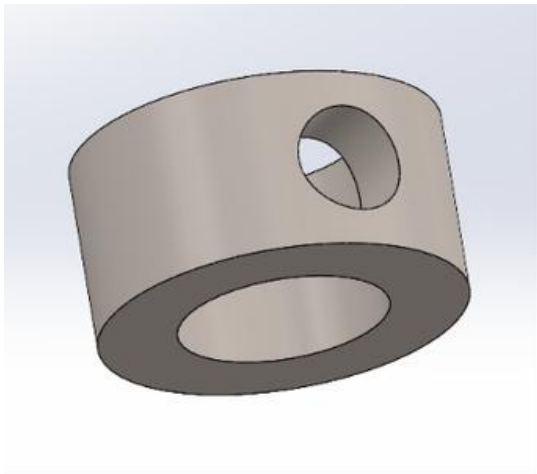
**Material justification:** Used for joining structural components because it requires good strength while maintaining low weight. 6061-T5 has better mechanical strength than many aluminum alloys and is easy to machine.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.002914 | 650          | 18.94         | 97.35             |

## Part 24:lock collar 8 mm

Material : alloy steel



**Mass : 5.26 grams**

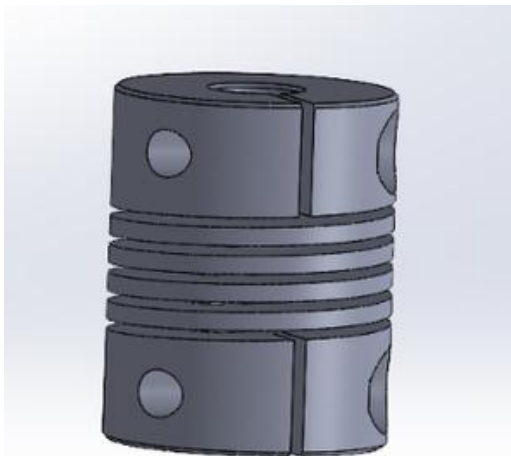
**Material justification:** A lock collar holds shafts/components in position and experiences clamping forces. Alloy steel is used because it provides high strength and resistance to deformation.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00526  | 300          | 1.58          | 53.95             |

## Part 25 : flexible coupler

Material : aluminum(body)



**Mass : 14.57 grams**

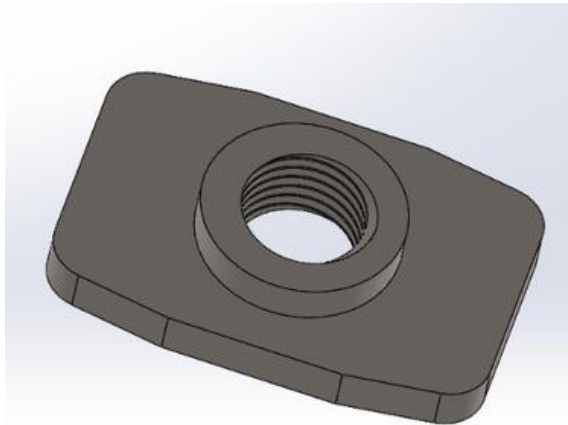
**Material justification:** The coupler connects rotating shafts and needs low inertia so that it can transmit motion efficiently. Aluminum provides a lightweight but sufficiently strong body.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.001457 | 600          | 8.74          | 71.85             |

## Part 26 : Tee nut

Material : Alloy steel



**Mass : 1.79 grams**

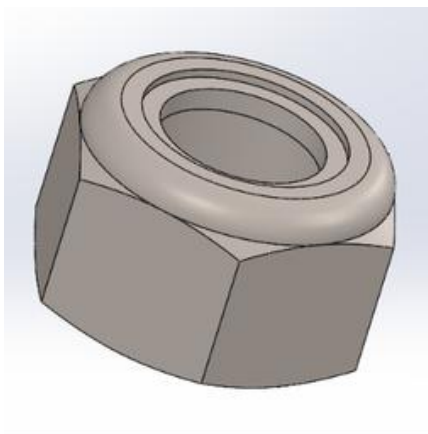
**Material justification:** Tee nuts are inserted into slots and tightened repeatedly, so they require high strength and resistance against thread damage. Alloy steel provides durability under mechanical loading.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00179  | 300          | .54           | 51.35             |

## Part 27 : Nylon insert lock nut

Material : alloy steel



**Mass : 1.33 grams**

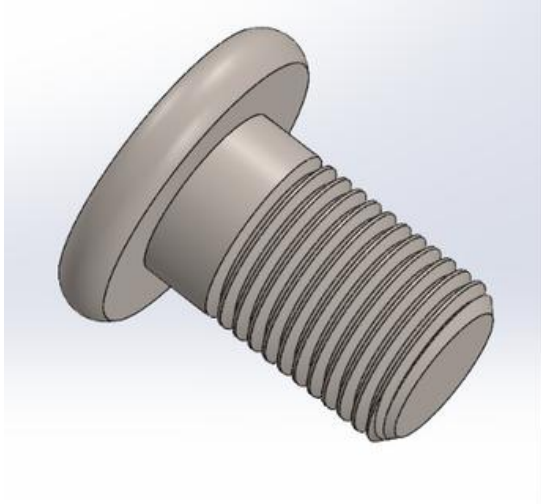
**Material justification:** The steel body provides mechanical strength, while the nylon insert prevents loosening due to vibration by creating friction with the bolt threads

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00133  | 300          | 0.40          | 51.00             |

## Part 28 : Low Profile Screw M5x8

Material : Alloy steel



**Mass : 1.67 grams**

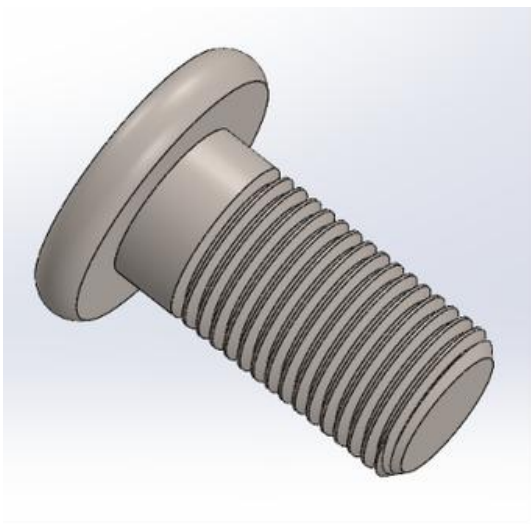
**Material justification:** M5×8 screws are made from alloy steel because they need to withstand tensile and shear forces during fastening. The material provides high strength and hardness, allowing the screw to handle tightening torque while maintaining a secure connection.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00167  | 300          | 0.50          | 51.25             |

## Part 29 : Low Profile Screw M5x10

Material : Alloy steel



**Mass : 1.94 grams**

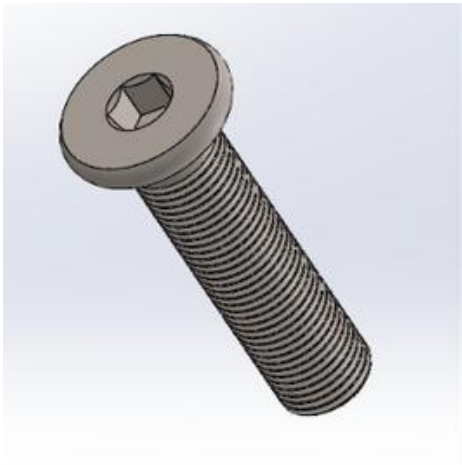
**Material justification:** M5×10 screws use alloy steel due to its excellent load-bearing capacity and resistance to deformation. The high strength and wear resistance of alloy steel ensure reliable fastening and durability under mechanical stress.

### Cost analysis in BDT

| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00194  | 300          | 0.58          | 51.45             |

**Part 30 : Low Profile Screw M5x15**

**Material : Alloy steel**



**Mass : 3.29 grams**

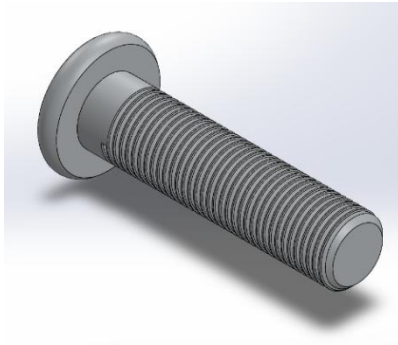
**Material justification:** M5x15 screws are manufactured from alloy steel because their longer length requires sufficient strength to resist bending, shear, and tensile loads. Alloy steel provides high toughness and fatigue resistance, ensuring stable and long-lasting fastening performance.

**Cost analysis in BDT**

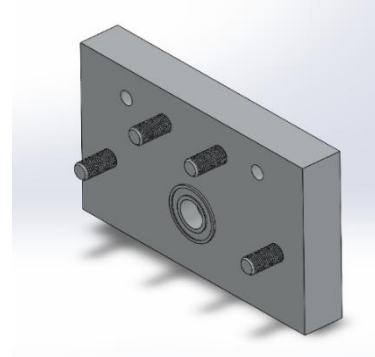
| Mass(kg) | Rate(BDT/kg) | Material cost | Total cost(Qty 1) |
|----------|--------------|---------------|-------------------|
| 0.00329  | 300          | 1.49          | 52.48             |

## Part 31: Low Profile Screw M5x20

Material: A286 Iron



Mass = 3.37  
grams



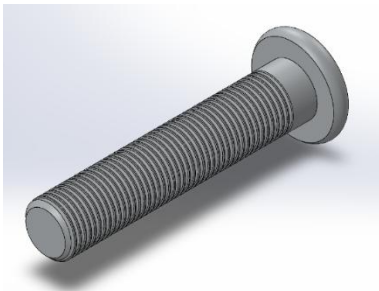
**Material Justification:** Prevents rusting and seizing when exposed to water-soluble coolants and cutting fluids.

### Cost Analysis (in BDT)

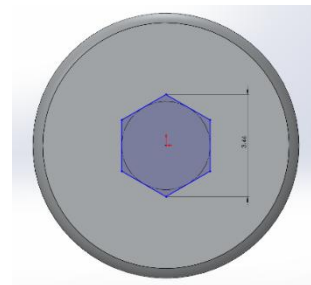
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 657.35       | 8.945         | 564.564        | 461.916       | 1035.425   |

## Part 32: Low Profile Screw M5x25

Material: A286 Iron



Mass = 4.07  
grams



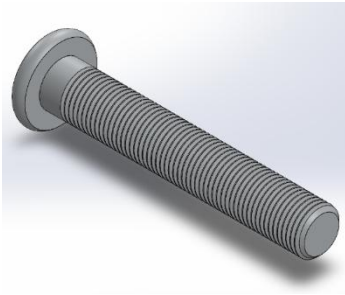
**Material Justification:** Prevents rusting and seizing when exposed to water-soluble coolants and cutting fluids.

### Cost Analysis (in BDT)

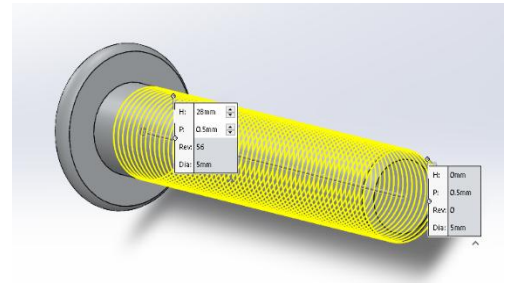
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 657.35       | 11.927        | 567.252        | 464.116       | 1043.300   |

### Part 33: Low Profile Screw M5x30

Material: A286 Iron



Mass= 4.76  
grams



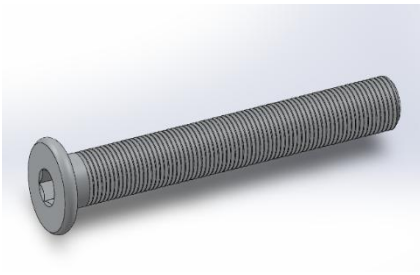
**Material Justification:** Prevents rusting and seizing when exposed to water-soluble coolants and cutting fluids.

#### Cost Analysis (in BDT)

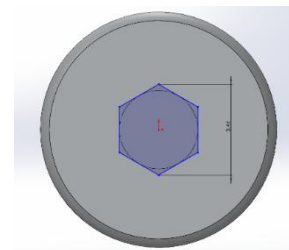
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 657.35       | 12.201        | 567.9245       | 464.666       | 1044.792   |

### Part 34: Low Profile Screw M5x40

Material: A286 Iron



Mass = 6.15  
grams



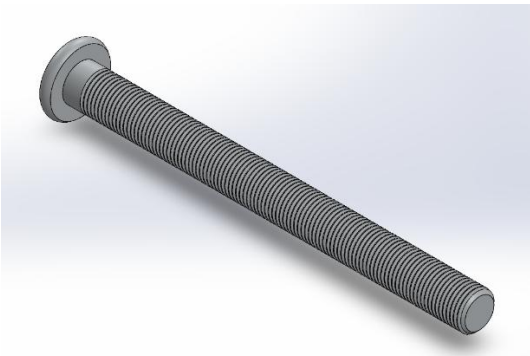
**Material Justification:** Prevents rusting and seizing when exposed to water-soluble coolants and cutting fluids.

#### Cost Analysis (in BDT)

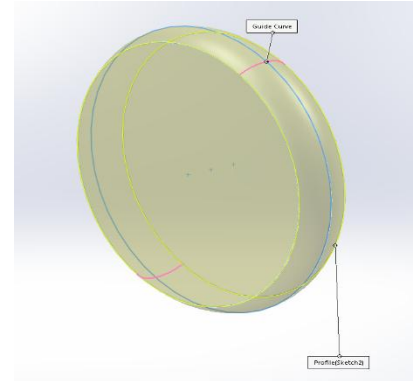
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 657.35       | 17.89         | 568.6          | 465.2154      | 1051.705   |

## Part 35: Low Profile Screw M5x55

Material: A286 Iron



Mass =  
8.24  
grams



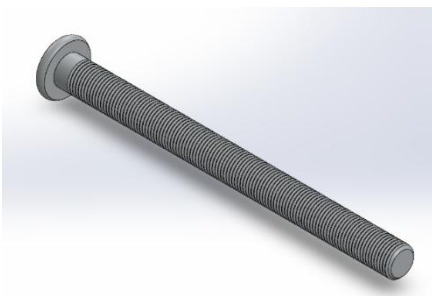
**Material Justification:** Prevents rusting and seizing when exposed to water-soluble coolants and cutting fluids.

### Cost Analysis (in BDT)

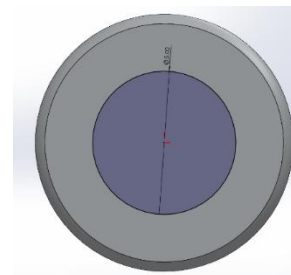
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 657.35       | 23.853        | 570.613        | 466.865       | 1061.331   |

## Part 36: Low Profile Screw M5x60

Material: A286 Iron



Mass = 8.93  
grams



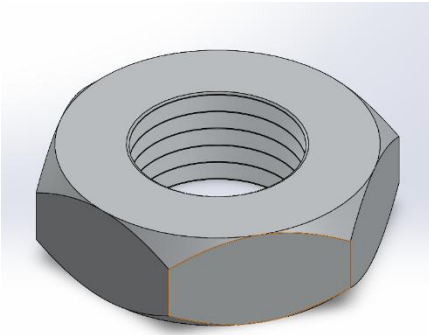
**Material Justification:** Prevents rusting and seizing when exposed to water-soluble coolants and cutting fluids.

### Cost Analysis (in BDT)

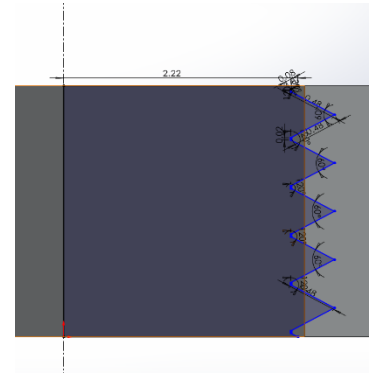
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 657.35       | 26.835        | 570.613        | 466.865       | 1064.313   |

## Part 37: Acme Anti Backlash - Nut

Material: PA Type 6



Mass =  
0.11 grams



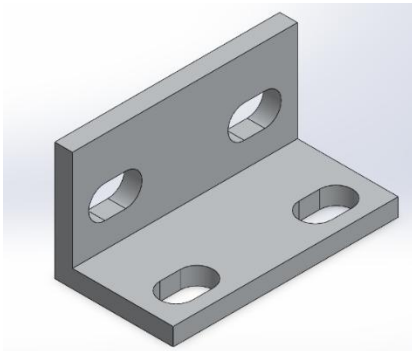
**Material Justification:** PA6 is an economical, self-lubricating polymer that eliminates stick-slip motion and thread wear under constant anti-backlash spring tension. Its high toughness and vibration-damping capabilities absorb motor chatter during direction reversals, ensuring smooth,

### Cost Analysis (in BDT)

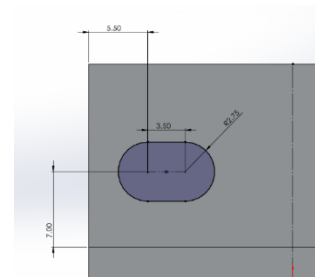
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 138.89       | 0.025         | 377.048        | 308.494       | 685.567    |

## Part 38: L bracket Double

Material: AISI 1020



Mass =  
29.25 grams



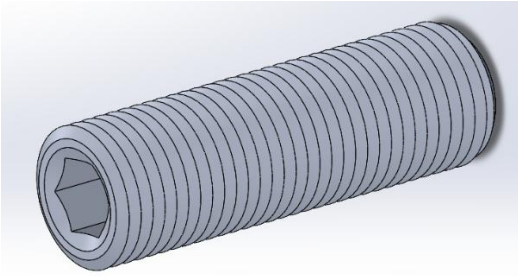
**Material Justification:** AISI 1020 mild steel provides high tensile strength and rigidity, preventing the double L-bracket from bending under heavy dynamic loads. It is also cheap, widely available in Bangladesh, and exceptionally easy to weld, drill, or bend into shape.

### Cost Analysis (in BDT)

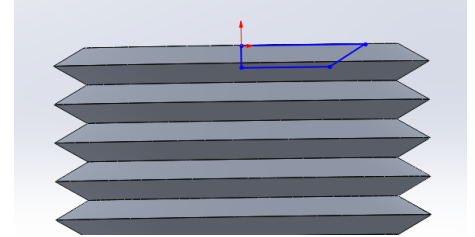
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 120          | 3.51          | 206.335        | 168.819       | 378.664    |

## Part 39: Acme Anti Backlash Nut

Material: PA Type 6



Mass =  
0.12  
grams



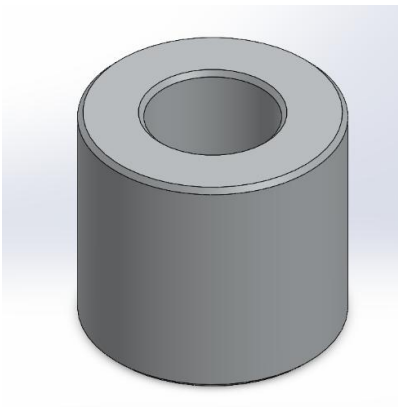
**Material Justification:** Highly economical, self-lubricating, and provides great vibration damping to eliminate stepper motor chatter.

### Cost Analysis (in BDT)

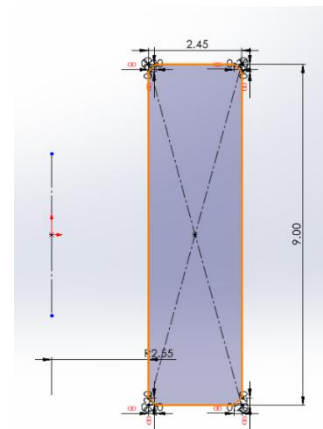
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 138.89       | 0.0167        | 376.376        | 307.944       | 684.337    |

## Part 40: 9 mm Aluminum spacer

Material: Aluminum



Mass = 1.41  
grams



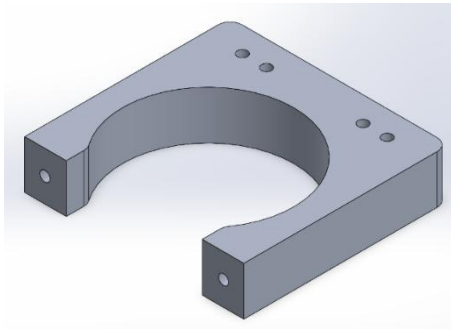
**Material Justification:** 6061-T6 aluminum is the optimal choice for spacers because its T6 heat treatment provides high yield strength), preventing deformation under heavy bolt clamping loads. It is lightweight, highly corrosion-resistant and exceptionally easy to machine to precise tight tolerances.

### Cost Analysis (in BDT)

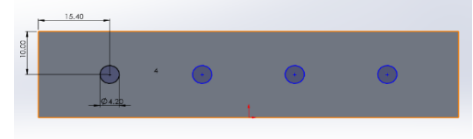
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 183.33       | 1.222         | 379.737        | 310.694       | 691.652    |

## Part 41: Spindle Part 1

Material: Aluminum



Mass= 200.11  
grams



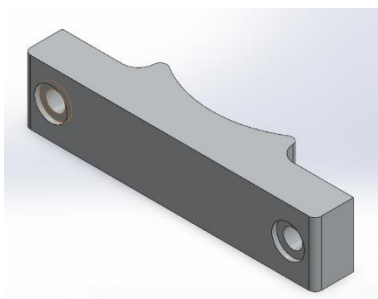
**Material Justification:** 6061-T6 aluminum is the optimal choice for spacers because its T6 heat treatment provides high yield strength), preventing deformation under heavy bolt clamping loads. It is lightweight, highly corrosion- resistantant exceptionally easy to machine to precise tight tolerances.

### Cost Analysis (in BDT)

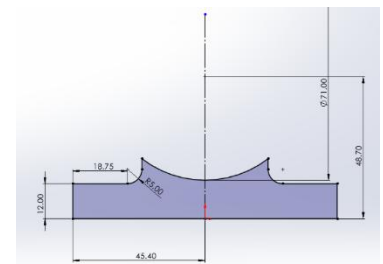
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 183.333      | 168.636       | 1117.030       | 913.934       | 2199.6     |

## Part 42: Spindle Part 2

Material: Aluminum



Mass= 65.01  
grams



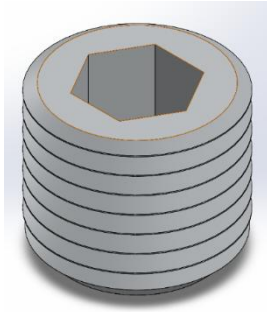
**Material Justification:** 6061-T6 aluminum is the optimal choice for spacers because its T6 heat treatment provides high yield strength), preventing deformation under heavy bolt clamping loads. It is lightweight, highly corrosion-resistantant exceptionally easy to machine to precise tight tolerances.

### Cost Analysis (in BDT)

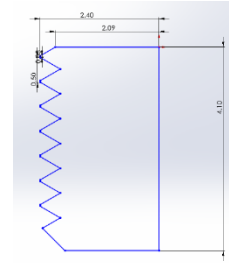
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 183.333      | 36.66         | 777.62         | 636.234       | 1450.514   |

## Part 43: Smaller Allen Screw

Material: AISI 304



Mass =  
0.13 grams



**Material Justification:** AISI 304 stainless steel is ideal for small Allen screws because its high tensile strength prevents the tiny hex socket from stripping under tightening torque. It also offers excellent corrosion resistance without needing extra plating, ensuring the threads won't rust or seize over time.

### Cost Analysis (in BDT)

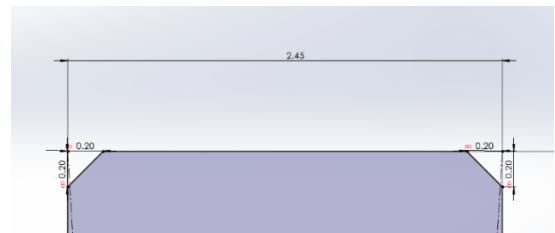
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 430          | 0.0559        | 376.376        | 307.944       | 684.376    |

## Part 44: 40 mm Aluminum spacer

Material: Aluminum



Mass =  
6.27 grams



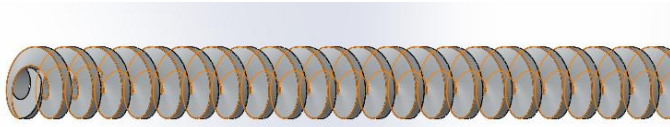
**Material Justification:** 6061-T6 aluminum is the optimal choice for spacers because its T6 heat treatment provides high yield strength, preventing deformation under heavy bolt clamping loads. It's lightweight, highly corrosion-resistant, and exceptionally easy to machine to precise tight tolerances.

### Cost Analysis (in BDT)

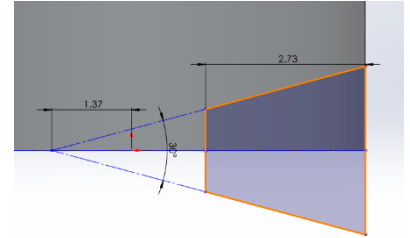
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 183.333      | 4.888         | 390.49         | 319.492       | 714.87     |

## Part 45: Lead screw 1000mm

Material: AISI 304



Mass = 204  
grams



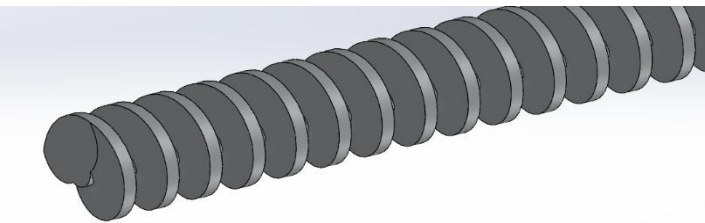
**Material Justification:** AISI 304 stainless steel is ideal for small Allen screws because its high tensile strength prevents the tiny hex socket from stripping under tightening torque. It also offers excellent corrosion resistance without needing extra plating, ensuring the threads won't rust or seize over time.

### Cost Analysis (in BDT)

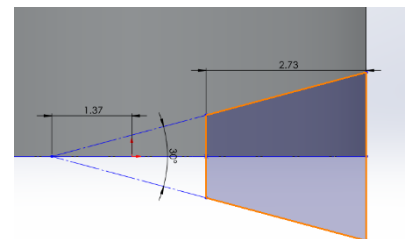
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 430          | 87.72         | 255.398        | 208.962       | 552.08     |

## Part 46: Lead screw 540mm

Material: AISI 304



Mass= 105.62  
grams



**Material Justification:** AISI 304 stainless steel is ideal for small Allen screws because its high tensile strength prevents the tiny hex socket from stripping under tightening torque. It also offers excellent corrosion resistance without needing extra plating, ensuring the threads won't rust or seize over time.

### Cost Analysis (in BDT)

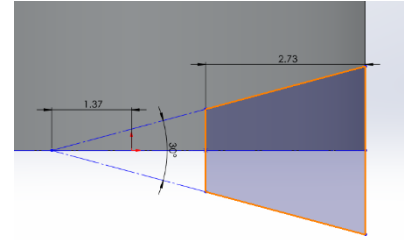
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 430          | 45.4166       | 411.325        | 336.5388      | 793.28     |

## Part 47: Lead screw 300mm

Material: AISI 304



Mass =  
56.40 grams



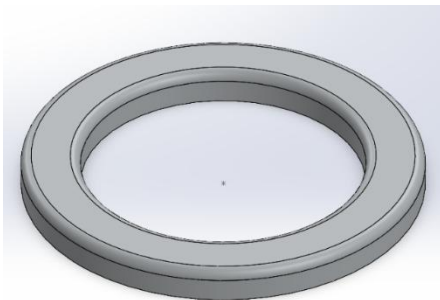
**Material Justification:** AISI 304 stainless steel is ideal for small Allen screws because its high tensile strength prevents the tiny hex socket from stripping under tightening torque. It also offers excellent corrosion resistance without needing extra plating, ensuring the threads won't rust or seize over time.

### Cost Analysis (in BDT)

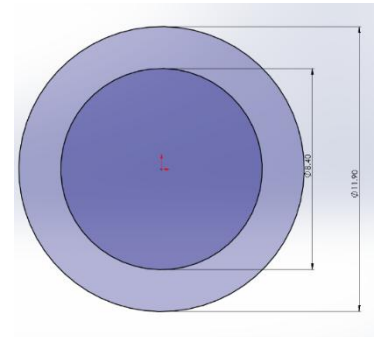
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 430          | 24.252        | 395.195        | 323.341       | 742.788    |

## Part 48: Shim 8mm

Material: AISI 304



Mass = 0.44  
grams



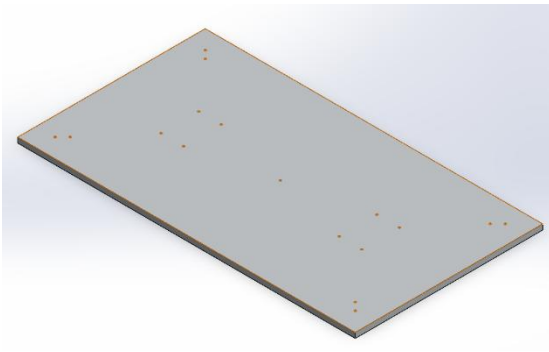
**Material Justification:** High wear resistance, maintains precise thickness under heavy load without compressing.

### Cost Analysis (in BDT)

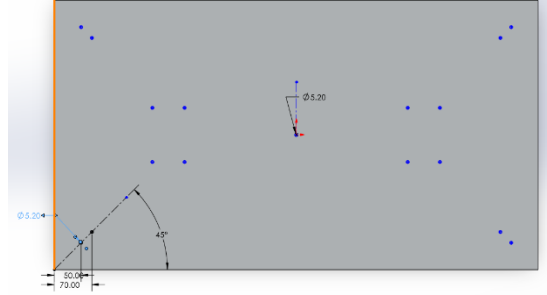
| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 430          | 0.1892        | 377.048        | 308.494       | 685.731    |

## Part 49: MDF Plate

Material: Aluminum



Mass =  
17000.00248  
grams

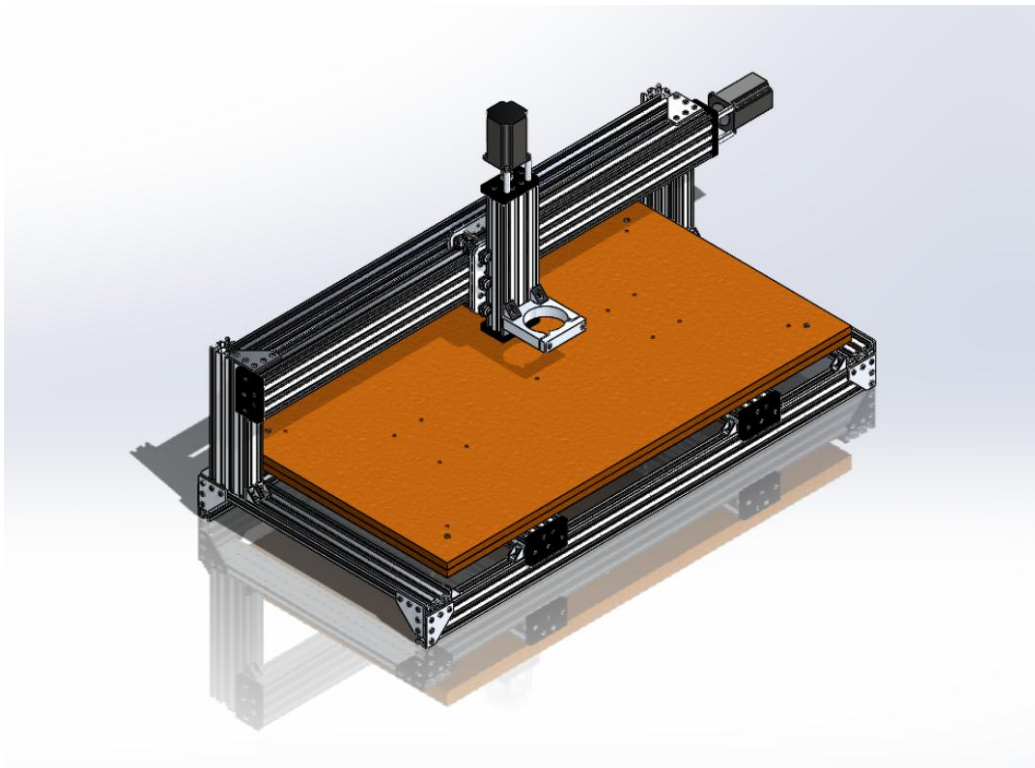


**Material Justification:** 6061-T6 aluminum is the optimal choice for spacers because its T6 heat treatment provides high yield strength), preventing deformation under heavy bolt clamping loads. It's lightweight, highly corrosion-resistantant exceptionally easy to machine to precise tight tolerances.

### Cost Analysis (in BDT)

| Rate (Tk/kg) | Material Cost | Machining Cost | Overhead Cost | Total Cost |
|--------------|---------------|----------------|---------------|------------|
| 183.333      | 6873.75       | 241.284        | 197.414       | 7312.448   |

## Final Assembly



**Figure:** CNC Machine

## **Bill of Materials (BOM) (In BDT)**

| <b>Sl. No</b> | <b>Parts Name</b>                 | <b>Price/piece</b> | <b>Qty.</b> | <b>Cost</b> |
|---------------|-----------------------------------|--------------------|-------------|-------------|
| 1             | V slot 20x60x1000                 | 648.01             | 2           | 1296.02     |
| 2             | Joining plate 90 degree           | 18.94              | 10          | 189.4       |
| 3             | Low Profile Screw 5x15            | 1.49               | 22          | 32.78       |
| 4             | Tee nut                           | 0.54               | 98          | 52.92       |
| 5             | Low Profile Screw M5x10           | 0.58               | 38          | 22.04       |
| 6             | Low Profile Screw M5x8            | 0.5                | 64          | 32          |
| 7             | Cast Corner Bracket (with nubs)   | 1                  | 18          | 18          |
| 8             | V slot 20x60x500                  | 324                | 2           | 648         |
| 9             | C beam 40x80x250                  | 302.07             | 3           | 906.21      |
| 10            | Cast Corner Bracket               | 0.96               | 8           | 7.68        |
| 11            | V slot 20x40x960                  | 435.44             | 1           | 435.44      |
| 12            | L bracket Double                  | 3.51               | 2           | 7.02        |
| 13            | MDF Plate                         | 378.12             | 2           | 756.24      |
| 14            | C beam 40x80x500                  | 604.14             | 2           | 1208.28     |
| 15            | Low Profile Screw M5x20           | 8.95               | 44          | 393.8       |
| 16            | Ball bearing 8x16x5               | 6.6                | 8           | 52.8        |
| 17            | C-beam end mount                  | 83.80              | 8           | 670.4       |
| 18            | Nema 23 high torque stepper motor | 4900               | 4           | 19600       |
| 19            | Low Profile Screw M5x55           | 23.85              | 8           | 190.8       |
| 20            | 40 mm Aluminum spacer             | 4.888              | 8           | 39.1        |
| 21            | Flexible coupler                  | 8.74               | 4           | 34.96       |
| 22            | Lead screw 540mm                  | 45.42              | 2           | 90.84       |
| 23            | Lock collar 8 mm                  | 1.58               | 8           | 12.64       |
| 24            | Smaller allen screw               | 0.0559             | 8           | 0.4472      |
| 25            | Shim 8mm                          | 0.19               | 8           | 1.52        |
| 26            | C beam gantry plate xlarge        | 141.71             | 7           | 991.97      |
| 27            | 8mm Acme nut block                | 6.93               | 6           | 41.58       |
| 28            | 3 mm Aluminum spacer              | 0.43               | 14          | 6.02        |
| 29            | Nylon insert lock nut             | 0.4                | 32          | 12.8        |
| 30            | Low Profile Screw M5x60           | 26.84              | 16          | 429.44      |
| 31            | 6 mm Aluminum spacer              | 0.56               | 15          | 8.4         |
| 32            | 9 mm Aluminum spacer              | 1.222              | 12          | 14.66       |
| 33            | V wheel                           | 3.39               | 30          | 101.7       |

| <b>Sl. No</b> | <b>Parts Name</b>             | <b>Price/piece</b> | <b>Qty.</b> | <b>Cost</b> |
|---------------|-------------------------------|--------------------|-------------|-------------|
| 34            | Ball bearing 5x16x5           | 8                  | 60          | 480         |
| 35            | Precision shim 10x5x1         | 0.42               | 42          | 17.64       |
| 36            | Mini Eccentric spacer 6mm     | 1                  | 15          | 15          |
| 37            | Low Profile Screw M5x30       | 12.21              | 4           | 48.84       |
| 38            | C beam 40x80x1000             | 1208.28            | 1           | 1208.28     |
| 39            | Lead screw 1000mm             | 87.72              | 1           | 87.72       |
| 40            | 20 mm Aluminum spacer         | 1.88               | 8           | 15.04       |
| 41            | Lead screw 300mm              | 24.25              | 1           | 24.25       |
| 42            | Low Profile Screw M5x25       | 11.93              | 6           | 71.58       |
| 43            | 8 mm Acme anti backlash block | 10.34              | 1           | 10.34       |
| 44            | Acme anti backlash nut        | 0.025              | 1           | 0.025       |
| 45            | Acme Anti Backlash - Nut      | 0.0167             | 1           | 0.0167      |
| 46            | Spindle part 1                | 168.636            | 1           | 168.64      |
| 47            | Spindle part 2                | 36.66              | 1           | 36.66       |
| 48            | Black Angle Corner Connector  | 4.86               | 4           | 19.44       |
| 49            | Low Profile Screw M5x40       | 17.89              | 22          | 393.58      |
|               |                               |                    | 683         | 30902.96    |